

Roger Waggoner

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Application Support Engineer / QA Analyst | Platform Troubleshooting | SQL & APIs

PROFESSIONAL SUMMARY

Application support and QA analyst who independently develops and operates a high-traffic community and file-hosting platform. Full-stack troubleshooting experience across REST APIs, SQL/PostgreSQL data, Linux services, deployments, monitoring, and user-reported defects. Strong at reproducing issues, tracing behavior through logs/data/configuration, validating fixes, and writing clear technical documentation.

TECHNICAL SKILLS

Support / QA	Issue triage, ticket-style investigation, scope isolation, bug reports, reproduction steps, expected/actual results, escalation-ready notes, regression testing, Playwright E2E testing, Jira-style workflows
Diagnostics	Log analysis, SQL/data checks, REST API debugging, configuration review, monitoring, technical documentation
Web & Data	TypeScript, JavaScript, PHP, Node.js, Express, React, HTML/CSS, SQL, PostgreSQL, relational database design
Systems & Tools	Linux administration, Podman, Docker, Cloudflare/CDN caching, Git, CI/CD, monitoring, automated backups, application operations
Additional Languages	Python, C/C++, C#

EXPERIENCE

Platform Engineer & Systems Administrator <i>Independent Project – Community and File-Hosting Platform</i>	2023–Present <i>Remote</i>
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- Solely develop and operate a public community and file-hosting platform across two production servers; recently supported more than 800,000 unique visitors, 30 million requests, and 10 TB of data transfer in 30 days.
- Own custom application code and infrastructure across multiple languages and stacks, including TypeScript/React, PHP, PostgreSQL, Linux, Podman/Docker, deployments, monitoring, backups, and CDN/cache configuration.
- Investigate user-reported and operational issues through logs, SQL queries, service configuration, deployment history, and application code; implement fixes and validate behavior in production.

Quality Assurance Engineer (Contract) <i>Artificer Studios</i>	Jun 2022–Aug 2023 <i>Remote</i>
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- Tested unreleased game builds across core gameplay, UI, progression, and stability; reproduced and documented defects with clear steps, expected vs. actual behavior, severity notes, and build/version context.
- Verified fixes across build updates and supported regression testing and release-readiness checks by tracking issue status and communicating concise technical findings.

SELECTED PROJECTS

- **Zill O'll Infinite Plus Translation** — Reverse-engineered PSP message formats and MIPS code; patched engine memory and text-size limits, debugged memory corruption with PPSSPP/Rizin, and built validation tooling for discovered runtime constraints.
- **MicroRules** — C# expression DSL that parses and compiles expressions into executable functions at runtime; includes examples, benchmarks, tests, and clear error behavior for edge cases.
- **OMF Parser** — Python parser/dumper for Relocatable Object Module Format (OMF) files based on the TIS specification; built around careful data parsing, validation, and standards compliance.

EDUCATION

University of Pittsburgh , Pittsburgh, PA — Computer Science coursework	2015–2017
Community College of Westmoreland County , Youngwood, PA — Associate of Science	2012–2014

Projects and technical notes: hk47196.github.io.